Ch-2

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Ques:-

1.) $\left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)\left(x^2+\frac{1}{x^2}\right)\left(x^4+\frac{1}{x^4}\right)$

$\Rightarrow \left(x^2-\frac{1}{x^2}\right)\left(x^2+\frac{1}{x^2}\right)\left(x^4+\frac{1}{x^4}\right)$

$\Rightarrow \left(x^4-\frac{1}{x^4}\right)\left(x^4+\frac{1}{x^4}\right)$

$\Rightarrow x^8 - \frac{1}{x^8}$

2.) If $x+\frac{1}{x}=6$, find

a. $x^2+\frac{1}{x^2}$

b. $x^4+\frac{1}{x^4}$

sol:

a. $\left(x+\frac{1}{x}\right)^2=(6)^2$

$x^2+\frac{1}{x^2}+2.\cancel{x}.\frac{1}{\cancel{x}}=36$

$x^2+\frac{1}{x^2}=36-2$

$x^2+\frac{1}{x^2}=34$

b. $\left(x^2+\frac{1}{x^2}\right)^2=(34)^2$

$(x^2)^2+\left(\frac{1}{x^2}\right)^2+2.\cancel{x^2}.\frac{1}{\cancel{x^2}}=1156$

$x^4+\frac{1}{x^4}+2=1156$

$x^4+\frac{1}{x^4}=1156-2$

$x^4+\frac{1}{x^4}=1154$

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Q. If $x+y=12$ and $xy=32$, then find x^2+y^2?

sol:

$(x+y)^2 = (12)^2$

$x^2+y^2+2.xy = 144$

$x^2+y^2+64 = 144$

$x^2+y^2 = 144-64$

$= 80$

Q. Prove that $2a^2+2b^2+2c^2-2ab-2bc-2ac = (a-b)^2+(c-a)^2+(c-a)^2$

sol: $(a^2\times a^2 + b^2\times b^2 + c^2 - \times c^2 - 2ab - 2bc - 2ac)$

LHS

$\Rightarrow (a^2+b^2-2ab)+(a^2+c^2-2ac)+(b^2+c^2-2bc)$

$(a-b)^2+(a-c)^2+(c-b)^2$

$\underset{LHS}{[(a-b)^2+(a-c)^2+(c-a)^2]} = \underset{RHS}{(a-b)^2+(b-c)^2+(c-a)^2}$

Q. If $x+\frac{1}{x}=7$, then $x^3+\frac{1}{x^3}=?$

$\left(x+\frac{1}{x}\right)^3 = (7)^3$

$\Rightarrow x^3+\frac{1}{x^3}+3(x)^2\left(\frac{1}{x}\right)+3(x)\left(\frac{1}{x}\right)^2 = 343$

$\Rightarrow x^3+\frac{1}{x^3}+\frac{3x^2}{x}+\frac{3x}{x^2} = 343$

$\Rightarrow x^3+\frac{1}{x^3}+3x+\frac{3}{x} = 343$

$\Rightarrow x^3+\frac{1}{x^3}+$

$\Rightarrow x^3+\frac{1}{x^3}+3x.\frac{1}{x}\left(x+\frac{1}{x}\right) = 343$

$\Rightarrow x^3+\frac{1}{x^3}+3(7) = 343$

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$58+2ab=42$

$ab=21$

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$x^3+\frac{1}{x^3} = 343-21$

$x^3+\frac{1}{x^3} = 322$

Q. If $a+b=10$, and $a^2+b^2=58$ then find $a^3+b^3=?$

sol: $(a+b)^3 = (10)^3$

$a^3+b^3+3a^2b+3ab^2 = 1000$

$a^3+b^3+3ab(a+b) = 1000$

$a^3+b^3+3ab(10) = 1000$

a^3+b^3

$(a)^3+(b)^3 = (10)^3$

$\Rightarrow (a+b)(a^2+b^2-ab) = 1000$

$\Rightarrow (10)(58-ab) = 1000$

$\Rightarrow 580-10ab = 1000$

$\Rightarrow -10ab = 1000-580$

$\Rightarrow -10ab = 420$

$\Rightarrow ab = \frac{420}{-10}$

$a^3+b^3 = (a+b)(a^2+b^2-ab)$

$= 10\times(58-ab)$

$= 10\times 58-ab$

$= 10\times 58-21$

$= 370$ Ans

$(a+b)^2 = a^2+b^2+2ab$

$(10)^2 = 58+2ab$

$100 = 58+2ab$

$100-58 = 2ab$

$ab = 42/2 = 21$

Spiral

Q. If $x + \frac{1}{x} = 3$, then $x^6 + \frac{1}{x^6} = ?$

$\left(x + \frac{1}{x}\right)^2 = 3^2$

$x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = 9$

$x^2 + \frac{1}{x^2} = 9 - 2$

$\bullet\ x^2 + \frac{1}{x^2} = 7$

$\left(x^2 + \frac{1}{x^2}\right)^3 = (7)^3$

$= (x^2)^3 + \left(\frac{1}{x^2}\right)^3 + 3 \cdot x^2 \cdot \frac{1}{x^2}\left(x^2 + \frac{1}{x^2}\right) = 343$

$= x^6 + \frac{1}{x^6} + 3(3) = 343$

$= x^6 + \frac{1}{x^6} = 343 - 9$

$= x^6 + \frac{1}{x^6} = 334$

$= x^6 + \frac{1}{x^6} + 3(7) = 343$

$= x^6 + \frac{1}{x^6} = 343 - 21$

$= 322$

Q. If $x + \frac{1}{x} = 6$, find $x^4 + \frac{1}{x^4} = ?$

sol: $\left(x + \frac{1}{x}\right)^2 = (6)^2$

$x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = 36$

$\bullet\ x^2 + \frac{1}{x^2} = 34$

$\bullet \left((x^2)^2 + \frac{1}{x^2}\right)^2 = (34)^2$

$\Rightarrow x^4 + \frac{1}{x^4} + 2 \cdot x^2 \cdot \frac{1}{x^2} = 1156$

$\bullet\ x^4 + \frac{1}{x^4} = 1154$

Q. If $x^2 + \frac{1}{x^2} = 27$, then find $x - \frac{1}{x} = ?$

sol: $\left(x - \frac{1}{x}\right)^2 = (27)^2$

$= x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} = 729$

$= x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x}$

$= 27 - 2$

$= 25$

$\bullet \left(x - \frac{1}{x}\right)^2 = 25$

$\bullet\ x - \frac{1}{x} = \sqrt{25} = 5$

Q. If $x^2 + \frac{1}{x^2} = 83$, find $x^3 - \frac{1}{x^3} = ?$

sol: $x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right) \times \left(x^2 + \frac{1}{x^2} + ab\right)$

$\Rightarrow \left(\quad \right) \times (83 + ab)$

$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x}$

$= 83 - 2$

$= 81$

$\cdot\ x - \frac{1}{x} = 9$

$\left(x - \frac{1}{x}\right)^3 = (9)^3$

$= x^3 - \frac{1}{x^3} - 3 \cdot x^2 \cdot \frac{1}{x} + 3 \cdot x \cdot \frac{1}{x^2} = 729$

$= x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x}\left(x - \frac{1}{x}\right) = 729$

$= x^3 - \frac{1}{x^3} - 3(9) = 729$

$x^3 - \frac{1}{x^3} - 27 = 729$

$729 + 27$

$x^3 - \frac{1}{x^3} = 756$

Spiral

Q. $(29)^3 - (11)^3$

$a^3 - b^3$

$\Rightarrow$

$= (29 - 11) \times [(29)^2 + (11)^2 - (29)(11)]$

$= (18) \times [841 + 121 - 3201]$

$= 18 \times (962 - 3201)$

$= 17316 - 3201$

Q. If $x^4 + \frac{1}{x^4} = 47$, find $x^3 + \frac{1}{x^3} = ?$

sol: $- \left(x^2 + \frac{1}{x^2}\right)^2$

$= (x^2)^2 + \frac{1}{(x^2)^2} + 2 \cdot x^2 \cdot \frac{1}{x^2}$

$= x^4 + \frac{1}{x^4} + 2$

$= 47 + 2$

$= 49$

$\cdot \left(x^2 + \frac{1}{x^2}\right)^2 = 49$

$\cdot\ x^2 + \frac{1}{x^2} = \sqrt{49} = 7$

$\cdot \left(x + \frac{1}{x}\right)^2$

$= x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x}$

$= 7 + 2$

$= 9$

$\Rightarrow \left(x + \frac{1}{x}\right) = \sqrt{9} = 3$

Spiral

$$\left(x+\frac{1}{x}\right)^3$$

$$= (x)^3 + \left(\frac{1}{x^3}\right) + 3.x.\frac{1}{x}\left(x+\frac{1}{x}\right)$$

$$= x^3 + \frac{1}{x^3} + 3(3) = 27$$

$$= x^3 + \frac{1}{x^3} + 9 = 27 - 9$$

$$= x^3 + \frac{1}{x^3} = 18$$

Q. If $x+y = -4$, then $x^3+y^3-12xy+64 = ?$

sol: $(x+y)^3 = (-4)^3$

$x^3+y^3+3x^2y+3xy^2 = -64$

$x^3+y^3+3xy(-4) = -64$

$x^3+y^3-12xy = -64$

$x^3+y^3-12xy+64 = 0$

Q. If $p = 2-a$, then $a^3+6ap+p^3 = ?$

$(p)^3 = (2-a)^3$

$p^3 = (2)^3-(a)^3-3\times 2a(2-a)$

$p^3 = 8-a^3-6ap$

$p^3 = 8-a^3-6ap$

$p^3+6ap+a^3 = 8$

Q. If $a^2+b^2+c^2 = 20$ and $a+b+c = 0$, find $ab+bc+ca = ?$

$(a+b+c)^2 = (0)^2$

$= a^2+b^2+c^2+2ab+2bc+2ac = 0$

$= 20 + 2(ab+bc+ca) = 0$

$= 2(ab+bc+ca) = -20$

$= ab+bc+ca = -10$

$(2-p)^3 + 6(2-p)(2-a) + (2-a)^3$

$8 - p^3 - 3ab(a+b)$

* $3^4 \rightarrow$ Exponent / Indices / Power
Base

$(x^3)^{1/2}$

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• Rationalise the denominator

$$\frac{1}{(\sqrt{6}+\sqrt{5})-\sqrt{11}} \times \frac{(\sqrt{6}+\sqrt{5})+(\sqrt{11})}{(\sqrt{6}+\sqrt{5})+(\sqrt{11})}$$

$$\frac{\sqrt{6}+\sqrt{5}+\sqrt{11}}{(\sqrt{6}+\sqrt{5})^2-(\sqrt{11})^2} = \frac{\sqrt{6}+\sqrt{5}+\sqrt{11}}{(\sqrt{6})^2+(\sqrt{5})^2+2\sqrt{6}\cdot\sqrt{5}}$$

$$= \frac{\sqrt{6}+\sqrt{5}+\sqrt{11}}{6+5+2\sqrt{30}} = \frac{(\sqrt{6}+\sqrt{5}+\sqrt{11})}{(2\sqrt{30})} \times \frac{\sqrt{30}}{\sqrt{30}}$$

$$= \frac{(\sqrt{6}+\sqrt{5}+\sqrt{11})\sqrt{30}}{2\times 30} = \frac{(\sqrt{6}+\sqrt{5}+\sqrt{11})\sqrt{30}}{60}$$

laws of Exponents / Indices / Powers for Real no.

$(a>0)$

* $\sqrt[n]{a^m} = (a^m)^{\frac{1}{n}} = (a)^{m/n} \rightarrow a^p$ ~~→ is a rational no.~~

Eg $\rightarrow \sqrt[3]{a^{1/4}} = (a^{\frac{1}{4}})^{\frac{1}{3}} = (a)^{\frac{1}{12}} \rightarrow$ Rational no.

1. $a^m \times a^n = a^{m+n}$

Eg $\rightarrow$ • $2^3 + 2^{\frac{1}{3}} = (2)^{3+\frac{1}{3}} = (2)^{\frac{10}{3}}$

• $2^1 + 2^{\frac{1}{2}} + 2^{\frac{1}{2}} = (2)^{3+\frac{1}{2}} = (2)^{\frac{7}{2}}$

2. $a^m \div a^n = a^{m-n}$, where $a \neq 0$

Eg $\rightarrow \frac{2^4}{2^3} = (2)^{4-3} = 2^1 = 2$

3. $a^0 = 1$

Prove: $\frac{a^p}{a^q} = a^{p-q} \Rightarrow \frac{a^p}{a^p} = (a)^{p-p} = \boxed{1 = a^0}$

Eg $\rightarrow 5^0 = 1$

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4.) $(a^p)^q = a^{p\times q}$

Eg $\rightarrow (2^3)^{\frac{1}{3}} = (2)^{\frac{3}{3}} = 2^1 = 2$

• $[(a^q)^r]^s = a^{q\times r\times s}$

5.) $a^{-p} = \frac{1}{a^p}$

Eg $\rightarrow 5^{-2} = \frac{1}{5^2} = \frac{1}{25}$

6.) $\left(\frac{a}{b}\right)^{-p} = \left(\frac{a^{-p}}{b^{-p}}\right) = \left(\frac{b}{a}\right)^p$

7.) $a^p \times b^p = (a\times b)^p$

Eg $\rightarrow 2^2 \times 3^2 = (2\times 3)^2 = (6)^2 = 36$

Ex:- 2.1

Simplify

a.) $2^{\frac{2}{3}} \cdot 2^{\frac{1}{3}}$ $\quad (a^p \times a^q = a^{p+q})$

$\Rightarrow (2)^{\frac{2}{3}+\frac{1}{3}} = (2)^{\frac{3}{3}} = 2^1 = 2$

b.) $\left[3^{\frac{1}{5}}\right]^4 = (3)^{\frac{4}{5}}$ $\quad (a^m)^n = a^{mn}$

c.) $\frac{7^{\frac{1}{5}}}{7^{\frac{1}{3}}} = (7)^{\frac{1}{5}-\frac{1}{3}} = (7)^{\frac{3-5}{15}} = (7)^{\frac{-2}{15}} = \frac{1}{(7)^{\frac{2}{15}}}$ $\quad \left(\frac{a^m}{a^n} = a^{m-n}\right)$

d.) $13^{\frac{1}{5}} \cdot 17^{\frac{1}{5}}$ $\quad (a^m \times b^m) = (a\times b)^m$

$\Rightarrow (13\times 17)^{\frac{1}{5}}$

$\Rightarrow (221)^{\frac{1}{5}}$

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